

Assignment 1

Issued: Friday 11th September, 2026

Due: Sunday 20th September, 2026

1. Solvability of Linear Regression

(40 points)

Consider linear regression with two input features. Each training sample can be written as $\mathbf{x}^{(i)} = [x_1^{(i)}, x_2^{(i)}]^\top \in \mathbb{R}^2$ with scalar output $y^{(i)}$. The prediction function is

$$\hat{y}^{(i)} = \mathbf{w}^\top \mathbf{x}^{(i)} + \beta = w_1 x_1^{(i)} + w_2 x_2^{(i)} + \beta,$$

where $\mathbf{w} = [w_1, w_2]^\top$ contains the regression weights and β is the intercept. For an exact fit, the prediction must satisfy $\hat{y}^{(i)} = y^{(i)}$ for every training sample.

For three training samples, define the design matrix A , parameter vector $\boldsymbol{\theta}$, and target vector $\mathbf{y}$ by

$$A = \begin{bmatrix} x_1^{(1)} & x_2^{(1)} & 1 \\ x_1^{(2)} & x_2^{(2)} & 1 \\ x_1^{(3)} & x_2^{(3)} & 1 \end{bmatrix}, \quad \boldsymbol{\theta} = \begin{bmatrix} w_1 \\ w_2 \\ \beta \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y^{(1)} \\ y^{(2)} \\ y^{(3)} \end{bmatrix}.$$

The exact-fitting problem is therefore the linear system $A\boldsymbol{\theta} = \mathbf{y}$. It may have no solution, exactly one solution, or infinitely many solutions. In geometric terms, the regression model $y = w_1 x_1 + w_2 x_2 + \beta$ is a plane in (x_1, x_2, y) -space, and exact fitting asks for a plane that contains all three training points.

Consider the following three regression data sets.

Data set	$(\mathbf{x}^{(1)}, y^{(1)})$	$(\mathbf{x}^{(2)}, y^{(2)})$	$(\mathbf{x}^{(3)}, y^{(3)})$
1	$((0, 0), 1)$	$((1, 0), 3)$	$((0, 1), 4)$
2	$((0, 0), 1)$	$((1, 0), 2)$	$((2, 0), 4)$
3	$((0, 0), 1)$	$((1, 0), 2)$	$((2, 0), 3)$

(a) (18 points) For each of the three data sets:

- Construct the design matrix A and target vector $\mathbf{y}$.
- Use Gaussian elimination to reduce the augmented matrix $[A \mid \mathbf{y}]$ to row-echelon or reduced row-echelon form. From the reduced system, determine whether $A\boldsymbol{\theta} = \mathbf{y}$ has no solution, exactly one solution, or infinitely many solutions.
- If solution(s) exists, find its complete set of solutions. (For infinitely many solutions, find the basis for the solution set.)

(b) (12 points) Recall that the null space of an $(m \times n)$ matrix A is defined by

$$\text{Null}(A) = \{\mathbf{z} \in \mathbb{R}^n : A\mathbf{z} = \mathbf{0}\}.$$

For the design matrix A of each data set in Part (a):

- i. Find $\text{Null}(A)$ by giving a basis for it. Recall that the basis of the zero subspace is the empty set.
 - ii. Explain how the null space is related to uniqueness of the exact regression parameters.
- (c) (10 points) When $A\boldsymbol{\theta} = \mathbf{y}$ has infinitely many solutions, let $\boldsymbol{\theta}_0$ be one particular solution. Show that every exact solution has the form

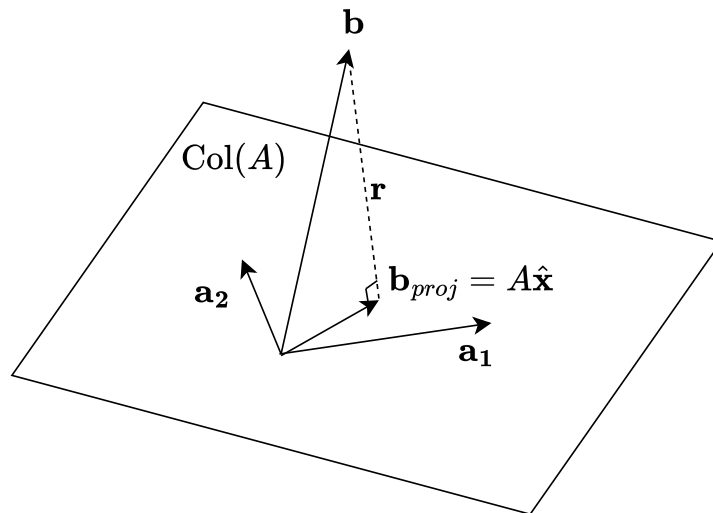
$$\boldsymbol{\theta} = \boldsymbol{\theta}_0 + \mathbf{z}, \quad \mathbf{z} \in \text{Null}(A).$$

Hint: Show that $\boldsymbol{\theta} - \boldsymbol{\theta}_0 \in \text{Null}(A)$.

2. The Pseudoinverse

(30 points)

If A is a square invertible matrix, then the system $A\mathbf{x} = \mathbf{b}$ has the unique solution $\mathbf{x} = A^{-1}\mathbf{b}$. For a rectangular matrix, however, an ordinary inverse does not exist. To obtain the closest approximate solution, we project $\mathbf{b}$ onto $\text{Col}(A)$ and choose $\hat{\mathbf{x}}$ such that $A\hat{\mathbf{x}}$ equals this projection.



Assume first that $A \in \mathbb{R}^{m \times n}$ has linearly independent columns, so that $A^T A$ is invertible.

- (a) (15 points) Show geometrically that

$$\hat{\mathbf{x}} = (A^T A)^{-1} A^T \mathbf{b}.$$

Here $A^+ = (A^T A)^{-1} A^T$ is the Moore–Penrose pseudoinverse of A .

Hint: Show that the residual $\mathbf{r} = \mathbf{b} - A\hat{\mathbf{x}}$ satisfies $A^T \mathbf{r} = \mathbf{0}$.

- (b) (15 points) Now let A be arbitrary, possibly with linearly dependent columns. Recall that the Euclidean inner product and norm are defined by

$$\|\mathbf{u}\|_2 = \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle} = \sqrt{\mathbf{u}^T \mathbf{u}}.$$

The least-squares problem is

$$\min_{\mathbf{x}} \|A\mathbf{x} - \mathbf{b}\|_2.$$

Show that $\hat{\mathbf{x}} = A^+\mathbf{b}$ is a least-squares solution. Then show that it is the unique least-squares solution of minimum Euclidean norm; that is,

$$\|\hat{\mathbf{x}}\|_2 \leq \|\mathbf{x}\|_2$$

for every least-squares solution $\mathbf{x}$, with equality only when $\mathbf{x} = \hat{\mathbf{x}}$.

Hint 1: Use the fact that $AA^+\mathbf{b}$ is the orthogonal projection of $\mathbf{b}$ onto $\text{Col}(A)$.

Hint 2: As in Problem 1(c), write every least-squares solution as $\mathbf{x} = \hat{\mathbf{x}} + \mathbf{z}$, where $\mathbf{z} \in \text{Null}(A)$.

Remark. When a regression problem has many solutions, all of those solutions make the same predictions on the training data. The minimum-norm solution is often preferred because it uses the smallest overall parameter magnitude. It avoids adding arbitrary components from $\text{Null}(A)$ that do not improve the fit. This can provide a simpler and more stable model and may improve generalization to new data.

If A is rank deficient, $A^\top A$ is not invertible, but the Moore–Penrose pseudoinverse A^+ is still defined and selects the minimum-norm least-squares solution.

3. Chain Rule

(10 points)

Suppose I have a function $q : \mathbb{R}^d \rightarrow \mathbb{R}$ defined as

$$q(x) = f(g(x)), \quad x \in \mathbb{R}^d,$$

where $g : \mathbb{R}^d \rightarrow \mathbb{R}$ is defined as $g(x) = \sum_{i=1}^d x_i$ (namely, it takes summation over all entries of x), and $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined as $f(y) = y^4$. Derive the expression of $\frac{\partial q(x)}{\partial x}$.

4. Chain Rule for Vector-valued Function

(20 points)

Suppose I have a scalar function q defined as

$$q(x; a, b) = b \cdot \phi(a^\top \phi(x)), \quad x \in \mathbb{R}^d.$$

In particular,

- The function $\phi(x) = \log(1 + e^x)$ operates componentwisely (namely, if x is a scalar, $\phi(x) = \log(1 + e^x)$ is a scalar; if x is a d -dimensional vector, $\phi(x) = (\log(1 + e^{x_1}), \dots, \log(1 + e^{x_d}))^\top$ is also a d -dimensional vector).
- $a \in \mathbb{R}^d$ and $b \in \mathbb{R}$ are weight parameters.

Answer the following questions:

- (8 points) For $x \in \mathbb{R}^d$, figure out the dimensions of $\phi(x)$, $a^\top \phi(x)$, $\phi(a^\top \phi(x))$, and $q(x; a, b)$, respectively.
- (12 points) Calculate the expressions of $\frac{\partial q(x; a, b)}{\partial x}$, $\frac{\partial q(x; a, b)}{\partial a}$, $\frac{\partial q(x; a, b)}{\partial b}$.